

National University of Computer and Emerging Sciences
FAST School of Computing
Fall 2022
CS3002 Information Security
BSCS-7H

QUIZ # 2 GRADED

Date: October 13, 2022

Time Allowed: 20 minutes

Maximum Marks: 10

Instructions: Attempt all 4 questions in sequence. Please write your answers briefly and precisely. Unnecessary or irrelevant details will not add anything to your credit. Each question carries 2.5 marks.

- 1) How many round keys are used in each variant of AES? What is the size of encryption key in each variant?

Solution

10, 12, and 14 round keys respectively ... For AES, key sizes are 128, 192, and 256 bits respectively

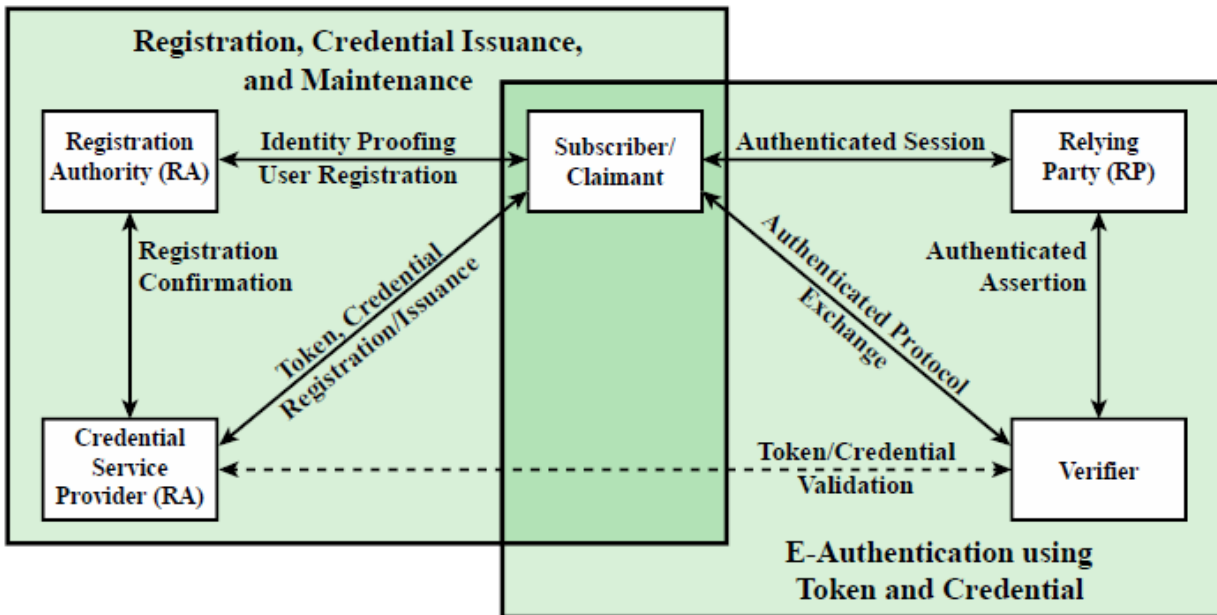
- 2) What is the concept of rainbow tables? Discuss

Solution:

A rainbow table is a listing of all possible plaintext permutations of encrypted passwords specific to a given hash algorithm. Rainbow tables are often used by password cracking software for network security attacks.

3) Show NIST's E-Authentication Architectural Model with the help of a labelled diagram.

Solution:



4) Discuss a limitation and an advantage of any two means of authentication.

Solution:

Passwords (something you know):

Advantage: convenience

Limitation: Memorization

Smart Cards (something you have):

Advantage: Can be given or shared with anyone

Limitation: Need to be kept in possession all the time. If lost then an issue.